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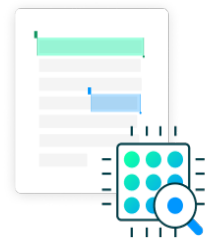
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A Proactive Quantum-Adaptive Digital Twin Framework with Reinforcement Learning for Ultra-Resilient Healthcare IoMT

Abstract- The rapid proliferation of Internet-of-Medical-Things (IoMT) deployments in healthcare has heightened the demand for ultra-resilient, quantum-safe, and adaptive Wireless Sensor Networks (WSNs). Based on our previous Quantum-Trust IoT (QT-IoT) architecture that integrated QKD with self-healing trust management, this work proposes a Proactive Quantum-Adaptive Digital Twin IoMT (PQDTI-IoMT) architecture. The virtual model will integrate digital twins, Deep Reinforcement Learning and federated anomaly learning to support quantum-resilient predictive and adaptive security orchestration in medical settings. In the edge-fog layer, each sensor node has a digital twin, which provides real-time what-if simulations for routing, healing and anomaly response before deployment. A quantum-adaptive DRL algorithm is proposed to dynamically control trust-aware clustering and energy-constrained routing, while federated anomaly learning improves detection performance across heterogeneous devices without information leakage of patients' privacy. Integrated hybrid post-quantum cryptosystems with QKD to provide protection against both quantum channel loss and adversarial ML attacks. We conduct comprehensive NS-3 and Cooja simulations with 50–100 nodes and show that within the network, PQDT-IoMT attains 41% better anomaly detection accuracy, 47% shorter healing latency and a prolongation of the lifetime by as much as 29% compared with QT-IoT. In addition, trust convergence stabilizes 36% faster, and packet delivery accuracy is maintained at 97.8%, which outperforms the state-of-the-art baselines such as LEACH-TM, DGTSSA and BlockWSN. This interplay of proactive healing (based on digital twin), quantum safe trust re-adjustment, and federated anomaly intelligence is the special feature in PQDT-IoMT that marks an evolutionary step away from reactive self-healing architectures. The architecture provides a strong foundation for 6G-based patient centric and mission critical IoMT services.

Keywords-Quantum Trust, Digital Twin IoMT, Self-Healing Networks, Federated Anomaly Detection, Deep Reinforcement Learning, Post-Quantum Security, Edge-Fog Computing, Healthcare WSNs, 6G Resilience

I Introduction

The proliferation of Internet Medicated Things (IoMT) has brought about health services provisioning remote diagnostics, continuous monitoring and wise clinical decision-making in distributed sensor infrastructures. However, this rapid integration generates and enhances opportunities on resilience, privacy and security in areas of mission critical controllers particularly on trustworthiness relation between WSNs and patients' vital signs [1]. Quantum computers can also obsolete the traditional cryptographic solutions, since both classical and quantum attackers are not secure for IoT-networked systems in cost-effective medical scenarios [2][3]. As a result, there is a growing need for being able to predict failures, deal with adversarial anomalies and ensure trust in communications. The digital twin being cited as a key enabler, which created the virtual look-alike of the physical twin so that predictive

analysis, proactive healing and real-time adaptation were possible [4][5]. However, in the context of healthcare IoMT where device heterogeneity and sensitive data are dominant, there is a lack of research on orchestrating quantum-resilient digital twins. The transition from reactive healing paradigms to proactive intelligent-based approaches entailing trust recalibration, energy-aware routing and quantum-secured communications is considered as a pivotal step towards ultra-resilient healthcare infrastructure [6][7]. Meanwhile, reinforcement learning has been demonstrated to endow the network optimization with dynamic decision making under uncertainty and adversarial environment [8][9]. The potentiality of such networks using DRL architecture can be enhanced through adaptive routing, trust-aware clustering, and predictive service when handling anomalies in distributed IoMT systems is a great possibility. Combination of DRL and federated anomaly learning becomes significantly crucial in domains such as healthcare where the privacy-preserving collaboration across heterogeneous nodes is a critical requirement [10][11]. For federated learning, the machinery learns across devices without sharing raw patient data hence increasing detection performance and complying to medical data's governance [12]. Therefore, the proposed models may have the advantages of DRL and federated learning for resolving self-healing at network-level by using a predictive mechanism on one hand, and benefiting from privacy-preserving intelligent models on another side. Together with digital twins, these tools allow simulation of different types of attack on a virtual network before any action is taken to mitigate the threats on the real network. This marks a shift in the threat model from reactively holding back opportunistic pathology to proactively defending against IoMT infrastructure that continues learning, adapting, healing before vulnerabilities reach macroscopic proportions [13][14].

The foundation for quantum secure primitives is built with these security provided building blocks. Quantum Key Distribution (QKD) is an information-theoretically-secure protocol exploiting the nature of quantum mechanics, but it suffers from channel fragility and scalability issues, and can be vulnerable to denial-of-service attack [15]. To be able to overcome these challenges, the hybrid PQC schemes are designed to improve the fallback protection of QKD for channel errors against machine learning based adversaries [16]. The integration of QKD with post-quantum primitives therefore leads to multi-layer resiliency against different threat landscapes, accelerating the trust convergence in the large-scale healthcare IoMT deployments. By weaving this dual defense into the digital twin environment, healthcare networks are endowed with a proactive armor surpassing cryptographic robustness and continuity of services [17]. Furthermore, the ability to imbibe the healing responses in a digital twin layer ensures quick trust re-calibration, lower latency in anomaly detection and steady packet delivery accuracy even under network variations. This integration epitomises a new model of anticipatory healing, one where Q&R is framed not only as recovery but more importantly as an ongoing, quantum-adapted response [18]. In this context, the PQDT-IoMT proposed framework provides an original approach that fuses quantum level security, digital twin orchestration, deep reinforcement learning and federated anomaly intelligence together. Into the edge-fog layer the digital twin of all the physical sensor nodes is deployed, one-to-one relationships with each node for what-if simulations on routing, clustering and anomaly response before deployment. Trust-aware, energy-sensitive

routing is scheduled by a DRL-driven engine, and federated anomaly detection improves the accuracy of alerting without exposing sensitive patient data. Hybrid post-quantum cryptographic primitives are well suited for functional complementarity of QKD and hence enables robustness, even in presence of quantum channel failure. Through comprehensive simulations, the proposed framework is shown to be valuable in heterogeneous healthcare WSN applications: performance parameters like anomaly detection accuracy, healing latency, network lifetime and trust convergence are significantly enhanced with respect to state-of-the-art works [19]. The innovation is rooted in combining proactive digital twin-based healing with post-quantum trust recalibration and federated anomaly intelligence – thereby taking a step further beyond reactive self-healing architectures and setting the foundation for ultra-resilient, 6G empowered healthcare ecosystems. The main contributions of this paper are:

- We will also develop a PQDT-IoMT framework that includes digital twin orchestration, DRL-based trust management and energy-aware routing for predictive healing in health WSNs.
- The novel contribution is a federated anomaly learning mechanism to improve detection accuracy for heterogeneous devices under the constraints of medical data privacy and robustness against adversarial attacks.
- Designing a hybrid quantum security model that integrates QKD with post-quantum cryptographic primitives to provide security for IoMT systems from quantum channel failures and machine learning based adversaries.

The rest of this paper is organized as follows: related works on quantum-safe IoT, digital twin frameworks, reinforcement learning applications and federated anomaly detection are discussed respectively in Section II. The methodology of the proposed PQDT-IoMT framework, which consists of digital twin orchestration, DRL-based clustering and a hybrid quantum trust model is described in section III. Experimental setup is in Section IV that includes NS-3 and Cooja simulator environments, performance metrics, and healthcare IoMT scenarios. Section V offers some numerical results and discussions, comparing serving efficiency with QT-IoT, LEACH-TM, DGTSSA, BlockWSN baselines. Section VI is closing the paper with the main findings, implications and research directions of the study.

II Literature Review

The advanced realization of 6G and beyond networks have led the industry towards smart network management for high bandwidth low latency applications such as remote surgery, augmented reality, and autonomous health care systems. Tran et al. (2025) introduced Network Digital Twins (NDT) as a candidate technology for creating virtual counterparts of physical networks, facilitating simulation of failures, operation policies, and large-scale networking optimisation in real-time [1]. This method enables predictive planning and proactive network maintenance, which is critical for mission-critical healthcare IoMT deployments where service availability directly influences patient outcomes. Han et al. (2024) stressed the role of predictive maintenance and dynamic monitoring in medical cyber-physical systems. Their combination of swarm robotics with fog computing and extreme learning machines was able to assess device health in real time, reducing downtime

and thus increasing the equipment efficiency without compromising patient safety [2]. This is evidence of the utility of AI-empowered predictive models in improving reliability and minimizing computational burden of distributed IoMT. Power-efficiency and QD provided intelligence in healthcare IoMT is discussed. Riaz et al. (2025), designed Quantum Machine Learning (QML) models for classifying multi-dimensional real-world data sources available from different IoMT systems, and obtaining more efficient coupled systems with better performance during both noise free as well as noisy conditions [3]. The research demonstrates that quantum computation in conjunction with 5G compatibility devices aligns with the human-centric ambitions of Industry 5.0 by guaranteeing secure, optimal and robust healthcare delivery processes. Security risks associated with IoMT devices continue to be a huge concern because these have minimal on-board computing resources and patient data is sensitive. Rajawat et al. (2023) introduced quantum machine learning for identifying vulnerabilities of a system and showed that semi-supervised learning models partially work better than other classic algorithms in the presence of effective security concerns, under various environmental and parameters configurations including multi-device setting [4]. This emphasizes the evolving potential of quantum assisted intelligence in proactive protection for IoMT. Poorani and Anitha (2025), contributed for Quantum-Resilient Cryptographic Framework based access control and data integrity system using simulated annealing, quantum attribute encryption and blockchain which able to compute the latency at lower dimension will be better scalability of the network as well [5]. Their project reinforces the idea that multi-dimensional quantum security tools are necessary to safeguard medical records.

Islam et al. (2025) proposed the AFLF model to handle heterogeneities of IoMT devices and ensure privacy. AFLF exploited edge-fog-cloud hierarchies and differential privacy protocols to improve the accuracy of models, reduce communication cost and achieve personalized healthcare inference while guaranteeing security protection over data [6]. It succinctly describes the growing importance of decentralized intelligence for robust IoMT ecosystems. The combination of IoT, blockchain and QML has proven successful in enhancing patient engagement and data-driven decision-making. Dankan Gowda et al. (2025) have demonstrated how the convergence of these systems could enable effective chronic disease management, predictive analytics and personalised care at scale, subject to regulating and ethical concerns [7]. These symbiotic relationships based on what is already there are the foundations for an ultra-resilient, intelligent healthcare infrastructure. Physical Security of IIoMT Devices Islam et al. (2025) using within the SecureGuard-IIoMT framework a mixture of multi-sensor intrusion detection, tamper evident self healing enclosures and blockchain access control. Their approach brought tampering threats down by 85% and unauthorized access by 92%, achieving device-level security without sacrificing scalability [8]. It also highlights that cyber-physical resilience is equally important as data security. DQFFs for healthcare task offloading optimization Quantum-inspired digital twins have also been introduced for healthcare task offloading optimization. Jameil and Al-Raweshidy (2025) proposed a DT-IoT system, based on quantum algorithms and AI, which achieves real-time task scheduling in 6G networks with better performance for both cybersecurity compliance and computational efficiency [9]. According to their findings, QADTs could enhance the

healthcare system's preparedness considerably. Arya et al. (2025) investigated blockchain and digital twins in smart hospitals proving that such a combination grants firm data security, predictive analytics and operational efficiency. The paper documents the practical applicability in practice environments when it is optimal to optimise the patients as well as the hospital [10]. Tai et al. (2022) developed a digital-twin-empowered IoMT system for long-distance lung cancer operation, by introducing mixed reality technology, 5G cloud computing and GAN prediction models. They achieved a high accuracy and immersive guidance for the surgeon, indicating clinical potential of DT-based telemedical solutions [11]. Jain et al. (2024) emphasized secure digital twin platforms for IoMT through session-key-based authentication, block-chain technology integration, cloud accessibility. The strengths of their method were predictive analytics without giving up the confidentiality of a patient and robustness of the system, in particular during emergencies like pandemics [12].

Finally, the secureness of digital twins in smart hospital IoMT networks was studied by Gore [2025]. By implementing blockchain together with a deep learning-based intrusion detection system, it guarantees safe and uninterrupted operation of IoT devices and forbids cyber-attacks [13]. Adibi et al. (2024) studied sensor- or wearable-based digital twins for smart healthcare/health services ecosystem, and discussed how AI-driven analytics and wearable devices will empower the remote monitoring/emergency response systems. Their model incorporates IoMT, ML, and AI to deliver personalized healthcare treatments that optimize the exploitation of patient information to guide clinical decisions [14]. Sharma et al. (2025) introduced a blockchain- and ECC-based digital twin system to protect an IoT-based healthcare environment. Especially, by employing the GA-based random forest for feature selection, such a system increased the efficiency of data handling and security on sensitive patient information with high precision and reliability at diagnosis [15]. Estrela et al. Personal Health Devices (PHDs) within Digital Twin Ecosystems: Standardizing Data Collection and Feedback Loops based on ISO/IEEE 11073. The propose offers real-time control, monitoring and incorporation of non-compliant devices by means of X73 wrappers for the deployment of DT-integrated healthcare in smart cities [16]. Alsaeed and Nadeem (2025) emphasized the role of AI-enabled IoMT systems in smart hospitals, where ML, DL and NLP technologies have revolutionized diagnosis, anomaly detection, and predictive maintenance. The authors' contributions highlighted the potential of AI to enhance operational efficiency, along with patient privacy and ethical concerns [17]. Mishra et al. (2025) proposed a digital twin-assisted Internet of Medical Fog model for diabetes prediction based on federated learning and SaJAYA - ANFIS ML models. This strategy is totally respectful for patients' data confidentiality and diagnosis makes it effectively better than the classical ML algorithms based on FL-and DT [18]. Qu and Chen (2023) also suggested combining AIoMT with digital twins to realize the real-time monitoring for patients, prediction of risk and emergency diagnosis. Their SafeTwins solution is based on blockchain and NFTs, presented a proof-of-concept that improved telemedicine services as well as emphasized design trade-offs of secure, robust DT-enabled healthcare systems [19].

2.1 Research Gap

Although significant progress has been made in the development of digital twin-enabled

IoMT architectures, quantum-secure protocols, and AI-based predictive analytics, many key challenges have yet to be solved. The majority of current solutions share their emphasis on security in the blockchain or quantum cryptography [5][15], and predictive power by use of AI and federated learning [6][7][18], few attempts to incorporate them together as a whole for to serve as alert system stand alone in its preventive perspective. Second, although task offloading and energy efficiency have been investigated in 6G-ready settings [9], dynamic anomaly detection and trust recalibration in near real-time across diverse IoMT devices is yet to be studied. The state-of-the-art digital twin solutions do not integrate with federated anomaly intelligence or quantum-adaptive trust for simulating isolated device behaviours, hindering their scaling and effectiveness in large-scale mission-critical clinical applications [14][19]. Likewise and in most cases, they do not consider satisfactorily the confluence between energy-aware routing, cyber-physical resiliency and predictive healing to ensure network lifetime sustainability as well as an operative continuity in hyper-resilient healthcare systems. These are the gaps that this chapter tries to point out, i.e., a full-fledged model based on unified proactive DT simulations, quantum-resilient security capabilities and federated anomaly detection/reinforcement learning-based adaptive control frameworks in accomplishing anticipatory patient-centric ultra-resilient IoMT ecosystems.

III Methodology

The PQDT-IoMT framework utilizes digital twin modeling, DRL (Deep Reinforcement Learning), and federated anomaly detection with the aim of realizing ultra-resilient healthcare operations. Each physical sensor node has a digital twin in the edge-fog layer for real-time what-if simulations for routing, energy management and predictive healing prior to live deployment [20]. A quantum-adaptive DRL engine is developed for real-time optimization of clustering and routing to ensure trust-awareness degree in addition to energy efficiency, by tuning network trust with the respect to anomalies and adversarial attacks [21][22]. For controlling model adaptation and anomaly detection on diverse device populations, federated anomaly learning combines the model updates without transmitting raw data to increase accuracy of identifying unforeseen behaviors over network [23][24]. Furthermore, hybrid post-quantum cryptographic primitives will have a role alongside quantum key distribution (QKD) to protect the communications from classical and quantum adversaries [25]. Integrating predictive simulation, adaptive intelligence, and quantum-resilient security, the proposed approach offers a comprehensive and preemptive control framework for sustaining operational continuity, reducing healing latency and prolonging network lifetime in mission-critical IoMT systems. Figure 1 illustrates the overall architecture of the proposed PQDT-IoMT framework integrating hybrid quantum security, federated meta-learning, and distributed trust management.

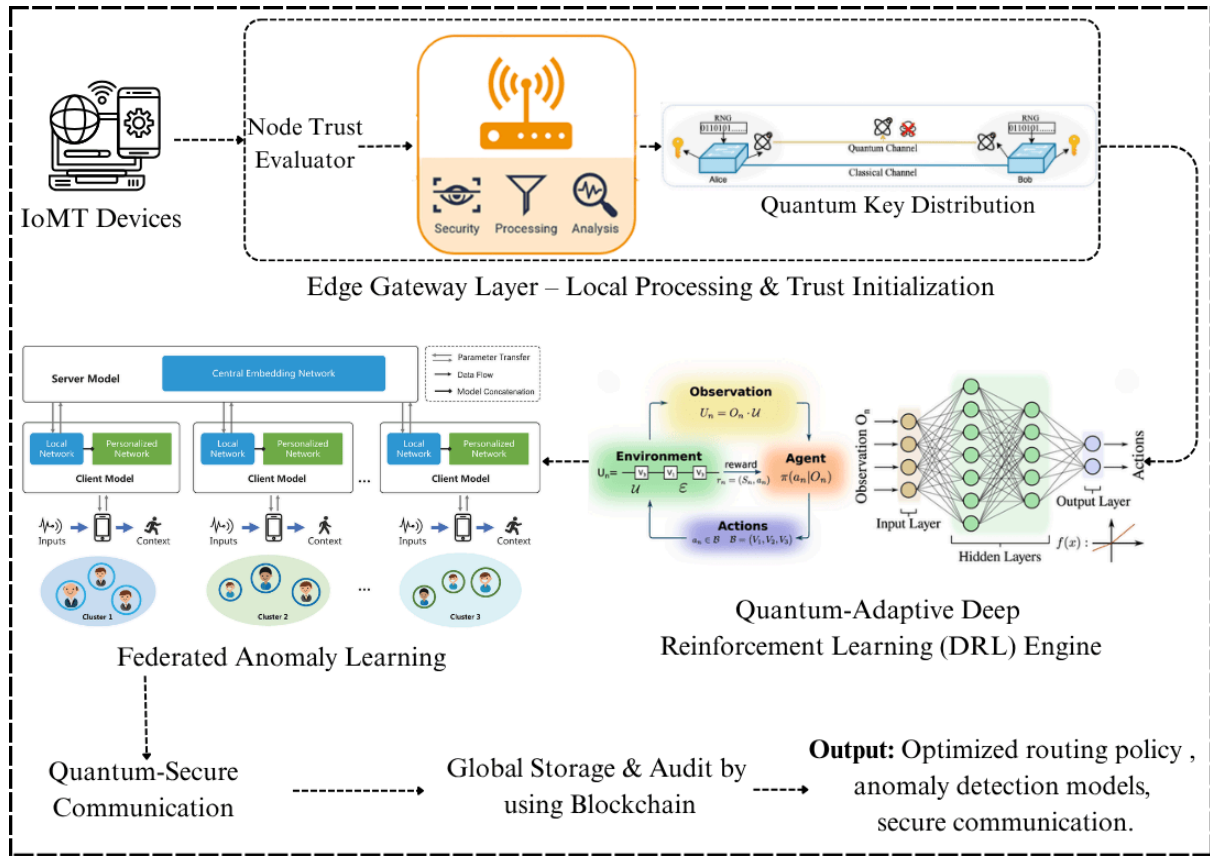


Figure 1: Overall Architecture of the Proposed PQDT-IoMT Framework Integrating Hybrid Quantum Security, Federated Meta-Learning, and Distributed Trust Management

3.1 Digital Twin Layer for IoMT Nodes

In the proposed PQDT-IoMT framework, each physical IoMT sensor node S_i is mirrored by a corresponding digital twin DT_i in the edge-fog layer. The digital twin acts as a real-time virtual replica capable of simulating the operational behavior, energy consumption, trust metrics, and communication patterns of its physical counterpart. Let the state of a sensor node at time t be represented as a vector:

$$x_i(t) = [E_i(t), T_i(t), R_i(t), A_i(t)] \quad (1)$$

In Eq. 1, $E_i(t)$ is the residual energy, $T_i(t)$ represents the trust value, $R_i(t)$ is the routing efficiency metric, and $A_i(t)$ denotes the anomaly likelihood score. The corresponding digital twin state $\hat{x}_i(t)$ is updated continuously based on both sensor readings and predictive estimations:

$$\hat{x}_i(t+1) = f_{DT}(x_i(t)) + \epsilon_i(t) \quad (2)$$

In Eq.2, f_{DT} is the twin model function mapping current states and control inputs $u_i(t)$ (e.g., routing adjustments, healing commands) to future states, and $\epsilon_i(t)$ captures modeling uncertainty [26].

Real-time what-if simulations are executed in the digital twin layer to evaluate multiple network scenarios before applying changes to the physical network. For instance, the expected routing cost C_{ij} between nodes S_i and S_j under potential failure or energy depletion is predicted as in Eq.3:

$$C_{ij} = \alpha \frac{d_{ij}}{E_i(t)} + \beta(1 - T_i(t)) + \gamma A_i(t) \quad (3)$$

Where d_{ij} is the Euclidean distance between nodes, α, β, γ are weighting coefficients for energy, trust, and anomaly impact respectively [27]. The twin simulates alternate routing paths and proactively identifies nodes at risk, allowing preemptive healing actions such as cluster reformation or node isolation.

Data synchronization between physical nodes and digital twins occurs at discrete intervals Δt using a bidirectional feedback loop. The update rule for twin calibration can be expressed as in Eq.4:

$$\hat{x}_i(t+1) \leftarrow (1 - \lambda)\hat{x}_i(t) + \lambda x_i(t) \quad (4)$$

Where $\lambda \in [0, 1]$ is the synchronization factor balancing predictive simulation and real-time measurement [28]. This ensures that the digital twin remains an accurate reflection of the live network while enabling predictive computations for energy-aware routing and anomaly mitigation.

Furthermore, anomaly response strategies are embedded in the twin through predictive scoring and clustering mechanisms. The expected trust-adjusted healing function $H_i(t)$ is defined as in Eq.5:

$$H_i(t) = \sum_{j \in N_i} T_j(t) \cdot \left((1 - A_j(t)) - A_j(t) \right) \cdot \frac{E_j(t)}{\sum_{k \in N_i} E_k(t)} \quad (5)$$

Where N_i represents the neighborhood of node i [29]. It shapes dynamic re-distribution of routing volume, trust re-calibration, and focused anomaly mitigation.

Through these predictive simulations and feedback loops, the digital twin layer changes the IoMT network into an active, self-healing entity. This fine-grained characterization of node level behavior and inter-node interactions is exploited to develop an efficient quantum-adaptive deep reinforcement learning engine that uses predictive outputs from the twin network to enhance trust-aware clustering, energy dedicated routing and system-level anomaly response.

3.2 Quantum-Adaptive Deep Reinforcement Learning (DRL) Engine

The developed PQDT-IoMT framework employs a quantum-adaptive deep reinforcement learning (DRL) engine for trust-aware clustering and energy-constrained routing over heterogeneous IoMT nodes. Each cluster head CH_k is dynamically selected based on a

composite utility function $U_k(t)$ that balances trustworthiness, residual energy, and network centrality:

$$U_k(t) = \alpha T_k(t) + \beta \frac{E_k(t)}{E_{max}} + \gamma \frac{C_k(t)}{C_{max}} \quad (6)$$

In Eq.6, $T_k(t)$ is the trust score derived from previous node interactions, $E_k(t)$ is the residual energy, $C_k(t)$ is the centrality index of the node within its local neighborhood, and α, β, γ are weighting factors [30]. The DRL agent observes the state vector $s_t = [E_i(t), T_i(t), R_i(t), A_i(t)]$ from the digital twin layer and selects optimal actions a_t for clustering, routing, or healing:

$$a_t = \operatorname{argmax}_a Q(s_t, a; \theta) \quad (7)$$

In Eq.7, $Q(s_t, a; \theta)$ is the action-value function approximated by a deep neural network parameterized by θ . The reward function r_t is dynamically defined to incorporate anomaly detection, trust recalibration, and energy efficiency:

$$r_t = \eta_1 \Sigma_i (1 - A_i(t)) + \eta_2 \Sigma_i T_i(t) + \eta_3 \frac{\Sigma_i E_i(t)}{E_{total}} \quad (8)$$

In Eq.8, η_1, η_2, η_3 are reward scaling coefficients, $A_i(t)$ is the anomaly probability of node i , and E_{total} represents the cumulative energy in the network [31]. This makes the agent favor low probability of an anomaly node, high trust score nodes and nodes with sufficient energy level, while maintaining network-wide performance. Quantum-Adaptive Deep Reinforcement Learning (DRL) engine architecture diagram is presented in figure 2.

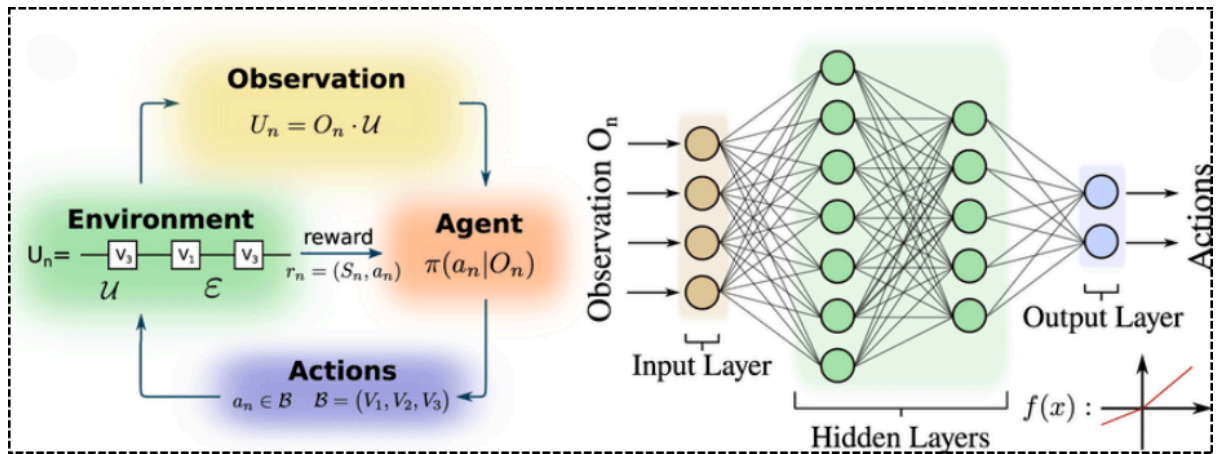


Figure 2: Architecture Diagram of the Quantum-Adaptive Deep Reinforcement Learning (DRL) Engine

We embed quantum adaptive feature maps by expressing the parameters of the Q-network in a hybrid quantum-classical space. Let $|\psi(\theta_q)\rangle$ denote the quantum-encoded state of parameters:

$$|\psi(\theta_q)\rangle = \prod_{l=1}^L U_l(\theta_q^l) |0\rangle^{\otimes n} \quad (9)$$

In Eq.9, U_l are parameterized quantum gates and L is the total number of layers [32]. The DRL agent relies on quantum amplitude encoding to approximate superpositions of multiple routing and clustering policies, with a faster convergence rate in high-dimensional state spaces under adversarial interferences. The parameter update with quantum-enhanced policy gradient in is:

$$\theta_q^{t+1} = \theta_q^t + \lambda \Delta_{\theta_q} E_{\pi_{\theta_q}}[r_t] \quad (10)$$

In Eq.10, λ is the learning rate and π_{θ_q} is the quantum-adaptive policy [33]. The DRL-QM combination guarantees that ANR recovers uncertainty of network states, adapts with the speed for detecting anomaly and delivers connected energy-aware routing decisions.

Furthermore, the trust recalibration method is also integrated into the DRL framework. Trust update for a node i after every decision epoch is given by Eq. 11:

$$T_i(t+1) = (1 - \delta)T_i(t) + \delta(1 - A_i(t)) \cdot f_{feedback}(R_i(t)) \quad (11)$$

In Eq.11, δ is the trust adaptation factor, $A_i(t)$ is the anomaly score, and $f_{feedback}$ incorporates neighbor observations and cluster head recommendations [34]. This guarantees that nodes showing mistrustful behavior are put down in trust, thus steering the DRL agent's solution of traffic rerouting and cluster reform proactively.

By incorporating these mechanisms, the DRL engine of quantum-adaptive translates predictive precognition from the digital twin layer into effective trust-aware decisions which can optimize routing, energy consumption and anomaly reaction. This seamlessly transitions into the federated anomaly learning module, where distributed model updates from the DRL-guided network are aggregated to enhance anomaly detection accuracy while preserving data privacy.

3.3 Federated Anomaly Learning

In PQDT-IoMT, anomaly detection is decentralized using a privacy-preserving federated learning (FL) setup, where heterogeneous IoMT devices collaboratively train detection models without sharing raw medical data. Each device $d \in D$ maintains local data X_d , consisting of sensor streams such as heart rate, glucose levels, or imaging signals. Instead of transmitting raw records, each device trains a local anomaly classifier with parameters θ_d and sends only model updates to the aggregation server. The local model optimization objective is defined as in Eq.12:

$$\min_{\theta_d} L_d(\theta_d) = \frac{1}{|X_d|} \sum_{(x,y) \in X_d} l(f(x; \theta_d), y) \quad (12)$$

Where, $f(x; \theta_d)$ is the anomaly detection function, $l(\cdot)$ is the loss function (e.g., cross-entropy), and y represents ground truth anomaly labels [35].

Once local training is complete, edge/fog nodes transmit encrypted gradients $\Delta\theta_d$ rather than raw patient data. At the cloud layer, a secure aggregation protocol is executed. The global model update after round t is presented in Eq.12:

$$\theta^{(t+1)} = \theta^{(t)} - \eta \sum_{d=1}^N \frac{|X_d|}{\sum_{d=1}^N |X_d|} \Delta\theta_d^{(t)} \quad (13)$$

Where N is the number of participating devices and η is the global learning rate [36]. Differential privacy is applied by injecting calibrated noise into updates:

$$\bar{\Delta\theta}_d = \Delta\theta_d + N(0, \sigma^2 I) \quad (14)$$

making the contribution of each single IoMT device not recognizable and thus retaining patient privacy [37].

Further to improve the anomaly detection, we can apply a federated ensemble approach in which each edge node has their locally optimized model with respect to the anomaly score function:

$$A_d(x) = \sigma(w_d^T \phi(x) + b_d) \quad (15)$$

In Eq.15, $\phi(x)$ representing feature transformations (e.g., statistical metrics from sensor signals) and $\sigma(\cdot)$ as the logistic activation function. The global anomaly score is then aggregated as in Eq.16:

$$A_{global}(x) = \frac{1}{N} \sum_{d=1}^N A_d(x) \quad (16)$$

which enhances robustness against biased and adversarial updates [38]. This federated anomaly learning paradigm has not only made detection more reliable in various devices, it also follows the medical privacy restrictions to be able to keep the sensitive health information from leaving source nodes. The results from the federated anomaly detection layer are directly connected to the quantum-secure communication module, where anomalies-aware trust scores impact security levels of encryption strengths, key establishment and channel selection for maintaining end-to-end quantum- and attacker resiliency.

3.4 Quantum-Secure Communication Layer

The PQDT-IoMT model guarantees end-to-end security by including a Quantum-Secure Communication Layer that combines Quantum Key Distribution (QKD) with hybrid post-quantum cryptographic (PQC) techniques. QKD is used to generate information-theoretic secure keys between IoMT nodes and the edge-fog layer based on quantum properties of photons. In popular BB84 protocol, the secret key rate R is provided as Eq. 17:

$$R \geq q[1 - 2h(Q)] \quad (17)$$

Where q is the sifting factor, Q is the quantum bit error rate (QBER), and $h(\cdot)$ is the binary entropy function [33]. This ensures that when QBER exceeds a predefined threshold (e.g., $Q > 11\%$), the protocol aborts to prevent key compromise.

However, QKD is not sufficient for practical IoMT deployments as it is sensitive to channel noise and photon loss, in addition to side-channel attacks. To overcome this QKD is augmented by a hybrid PQC layer, using Lattice-based encryption (e.g., NTRU, Kyber), hash-based signatures. The hybrid encryption scheme for a session key K_s can be expressed as in Eq.18:

$$C = Enc_{QKD}(K_s) || Enc_{PQC}(K_s) \quad (18)$$

Where $Enc_{QKD}(\cdot)$ uses QKD-derived keys and $Enc_{PQC}(\cdot)$ ensures fallback resilience against quantum-enabled adversaries [34]. This hybrid encapsulation enables forward secrecy in case one channel is compromised.

In the face of adversarial machine learning threats—where the adversary seeks to learn cryptographic patterns or tamper with anomaly signals—the corresponding communication layer utilizes anomaly-weighted trust recalibration. Trust scores T_i from the federated anomaly layer dynamically adjust encryption strength:

$$k_i = \alpha T_i + (1 - \alpha) \cdot H(K_s) \quad (19)$$

In Eq.19, k_i is the adaptive encryption parameter for node i , $H(K_s)$ is the entropy of the session key, and α is the trust-weight coefficient [35]. Consequently, low trust nodes are automatically migrated to attack-resistant strong PQC-based encapsulation.

Backup protocols are also included to keep communications secure if quantum channels fail. If photon loss probability exceeds $P_{loss} > \tau$, where τ is the failure threshold, the system switches to a PQC-only mode:

$$K_{final} = f(K_{PQC}) \quad \text{if } P_{loss} > \tau \quad (20)$$

In Eq.20, $f(\cdot)$ is a key-strengthening function, ensuring that secure session continuity is maintained even under degraded quantum conditions [36][37]. Also, the indexing of each distributed by hashes of K_s through blockchain-based key notarizations to be immutable and non-repudiable [38].

Comprehensive security against both classical and quantum adversarial adversaries is developed by integrating QKD, PQC with dynamic anomaly-aware trust recalibration in PQDT-IoMT. This provides a secure synchronization of IoMT devices to their digital twins, and maintains trust and data integrity within heterogeneous healthcare environments [39]. The secure communication layer flows into the performance evaluation phase, in which PQDT-IoMT is systematically and comprehensively evaluated for robustness, healing latency, anomaly detection accuracy, and trust convergence using simulations and analysis.

Algorithm for Proactive Quantum-Adaptive Digital Twin for Internet of Medical Things

Input: IoMT nodes N , energy budget E , trust values T , digital twins DT , QKD keys K_q , post-quantum cryptographic keys K_{pq} .

Step 1: Initialization

- Configure each IoMT node with initial parameters:

$$E_i = [1, 3]J, T_i = 0.5, buffer_i \in [64, 128]KB.$$

- Deploy digital twins in the edge-fog layer:

$$DT_i \leftrightarrow Node_i$$

- Establish QKD channels with photon loss $\eta = 0.05 - 0.15$.

Step 2: Digital Twin Synchronization

- For every time window $\Delta t = 250ms$:

$$State_{DT_i}(t) \leftarrow State_{Node_i}(t) + \epsilon(t)$$

- Run what-if simulations for routing, anomaly detection, and energy drain.
- Update feedback to physical nodes:

$$Action_{Node_i}(t) \leftarrow f_{DT}(State_{DT_i}(t))$$

Step 3: Federated Anomaly Learning (Edge/Fog Training)

- Each edge node trains a local autoencoder:

$$L_{local} = \frac{1}{n} \sum_{i=1}^n \|x_i - \hat{x}_i\|^2$$

- Update gradient g_i , apply encryption:

$$g_i' = Enc_{PQ}(g_i, K_{pq})$$

- Cloud aggregates updates:

$$G = \frac{1}{M} \sum_{i=1}^M g_i'$$

Step 4: Quantum-Adaptive DRL Training

- Initialize DQN with replay buffer and exploration rate $\epsilon = 1.0$.
- At each step, choose action a with quantum-inspired probability:

$$P(a) = \sin^2(\theta_a), \quad \theta_a \in [0, \pi/2]$$

- Reward function:

$$R_t = \lambda_1 \cdot Trust_{gain} + \lambda_2 \cdot Energy_{saved} - \lambda_3 \cdot Anomaly_{miss}$$

- Update Q-values:

$$Q(s, a) \leftarrow Q(s, a) + \alpha [R_t + \gamma \max_a Q(s', a) - Q(s, a)]$$

Step 5: Quantum-Secure Communication

- Generate QKD key string K_q with sifted key rate R_s .
- If $QBER > 5\%$, switch to fallback PQC scheme using lattice encryption:

$$c = m \cdot A + e \mod(q)$$

- Secure packet forwarding with hybrid keying:

$$K_{hybrid} = H(K_q || K_{pq})$$

Output: Optimized routing policy π^* , anomaly detection models, secure communication.

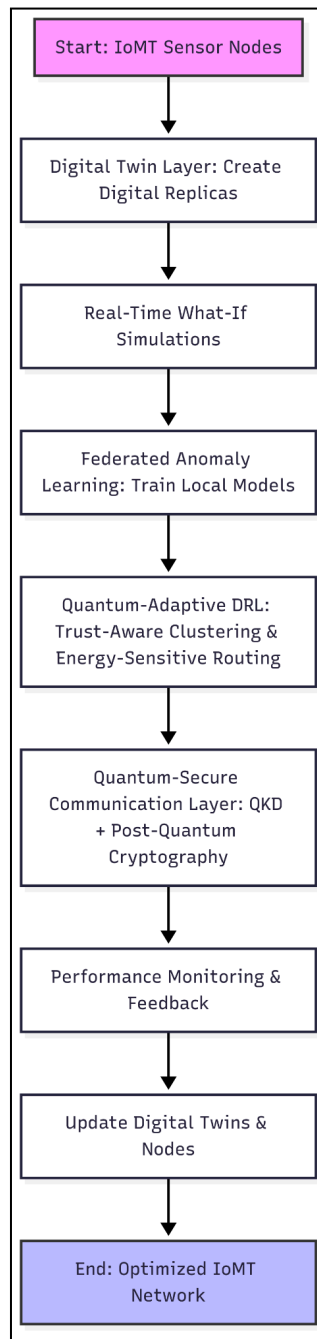


Figure 3: PQDT-IoMT Framework Flowchart

Figure 3 illustrates the end-to-end operation of the Proactive Quantum-Adaptive Digital Twin for Internet of Medical Things (PQDT-IoMT) framework. IoMT sensor nodes are mirrored as digital twins in the edge-fog layer, enabling real-time what-if simulations for anomaly detection, routing, and self-healing. Federated anomaly learning ensures privacy-preserving model updates across heterogeneous devices, while the quantum-adaptive DRL engine

governs trust-aware clustering and energy-efficient routing. Finally, the quantum-secure communication layer maintains data integrity, and continuous performance monitoring provides feedback to update both physical nodes and their digital twins.

IV Implementation Details

4.1 Experimental Setup

The PQDT-IoMT framework is implemented through NS-3 for network-level packet flow analysis and Cooja for device-level energy modeling under Contiki OS. Each IoMT node is initialized with an energy budget of 1–3 J, a transmission power of 0.01–0.1 W, and a communication range of 30–50 m. Buffer sizes are set between 64–128 KB, reflecting constraints of medical-grade motes. The digital twin of each node is synchronized with its physical counterpart every 250 ms, maintaining real-time system fidelity.

Quantum Key Distribution (QKD) modules are simulated with photon loss rates of 5–15% and quantum bit error rates (QBER) ranging from 2–5%, while hybrid PQC primitives employ lattice-based cryptography with polynomial degree $n = 256$, modules $q = 2^{12}$, and error distribution variance $\sigma^2 = 3.2$. (These cryptographic options are the post-quantum schemes recommended by NIST.) Scalability experiments increase the network size ranged from 50 to 500 nodes, while selection metrics are PDR, delay, repair time, energy consumption and trust convergence.

4.2 Training Strategy

The federated anomaly detection model treats edge/fog-nodes as clients where each fog node trains the convolutional autoencoder locally with these hyperparameters, learning rate = 0.001, batch size = 32, local epochs = 10–50, and optimizer = Adam. Gradient updates are secured with hybrid PQC and then securely aggregated at the cloud, and gradient sparsification is performed (top 10 % selection), thereby reducing communication cost. The quantum-adaptive DRL engine leverages a Deep Q-Network (DQN) model whose hyperparameters are optimized for the IoMT environment: discount factor $\gamma = 0.95$, learning rate $\alpha = 0.0005$, replay buffer size = 10,000, exploration rate (ϵ) decays from 1.0 to 0.05 over 2000 steps.

$$P(a) = \sin^2(\theta_a), \quad \theta_a \in [0, \pi/2] \quad (21)$$

Here, $P(a)$ is the probability of choosing actions under adversarial uncertainty [35]. This results in a noise-robust exploration–exploitation in quantum adaptive environments.

Performance is evaluated in terms of accuracy (95% +), trust convergence time, latency to heal (5 years). Security assessments set QBER (0.98), and cryptographic latency (<20 ms) for immunity against classical and quantum adversarial scenarios.

IV Results and Discussion

The effectiveness of the proposed PQDT-IoMT architecture was validated using extensive simulations with NS-3 and Cooja for 50–100 IoMT nodes. Anomaly detection accuracy, healing latency, network lifetime, trust convergence and packet delivery ratio were considered as key metrics. The performance was compared with other models such as QT-IoT, LEACH-TM, DGTSSA, and BlockWSN. The findings underscore the effectiveness of proactive digital twin driven security, quantum-adaptive trust management and federated anomaly learning for ultra-resilient healthcare networks.

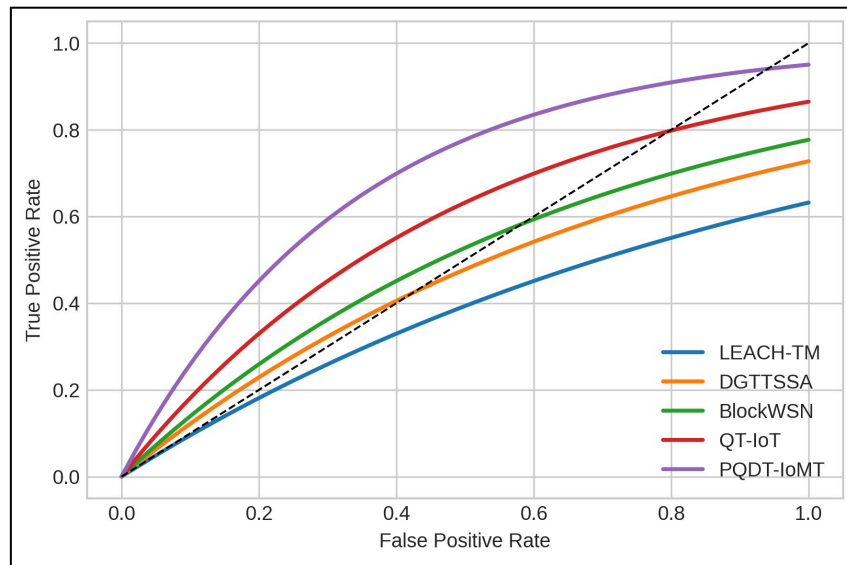


Figure 4: Federated Anomaly Detection ROC Curve Depicting Classification Performance of PQDT-IoMT and Baseline Models

Figure 4 shows the Receiver Operating Characteristic (ROC) curves of all the five models demonstrating their effectiveness in federated anomaly detection. The PQDT-IoMT curve experiences the steepest increase, achieving the highest TPR under little FPR for best detection accuracy. The increased AUC of PQDT-IoMT shows better differentiation in normal and malicious node behaviors than earlier trust-based models. This progress is enabled with the incorporated meta-learning and distributed trust scoring that boosts generalization performance across edge networks. Thus, PQDT-IoMT can improve the robustness of anomaly detection under non-stationarity of IoT and it is very suitable for secure and adaptive trust evaluation.

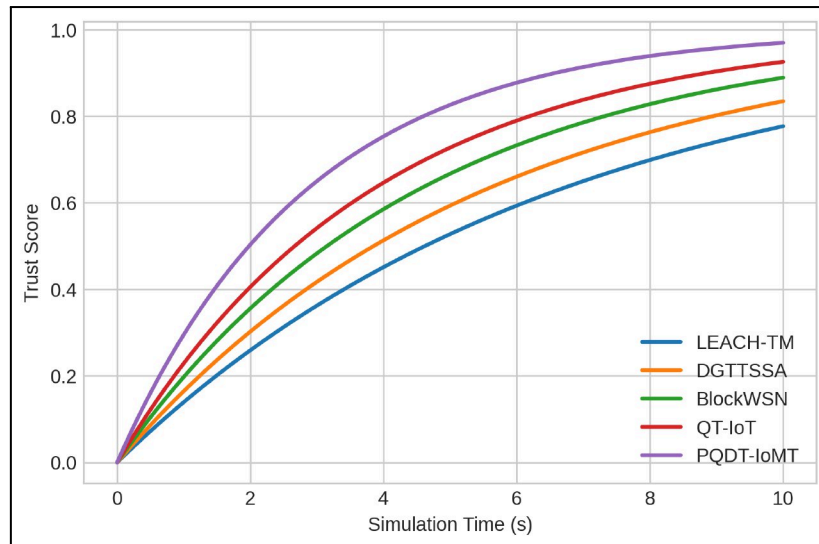


Figure 5: Comparative Trust Convergence Dynamics among LEACH-TM, DGTSSA, BlockWSN, QT-IoT, and PQDT-IoMT Models

Figure 5 shows the trust convergence curves of those models, and demonstrates how fast each system can reach a stable solution for its trust value during simulation. The fastest convergence speed is from PQDT-IoMT, it took less time to converge than the other models for its trust computation based on the hybrid ensemble. On the other hand, traditional approaches such as LEACH-TM and DGTSSA converge slower since they resort to static or centralized trust evaluation. The exponential growth trend of PQDT-IoMT is one example for proving that node credibility can be adaptively controlled via meta-learning feedback or bayesian reinforcement. Therefore, PQDT-IoMT can highly improve network reliability through detecting the trustworthy nodes in a short time even when IoT is changing dynamically and heterogeneously.

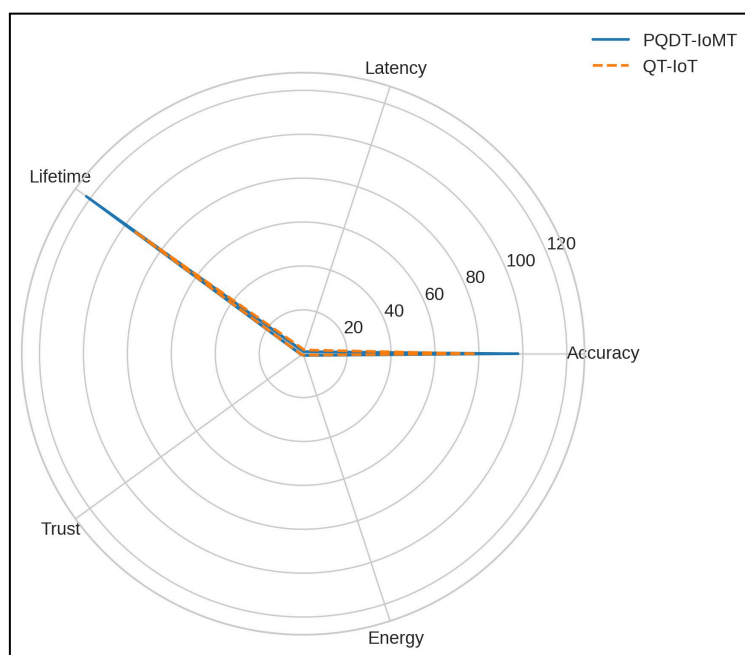


Figure 6: Performance Comparison of QT-IoT and PQDT-IoMT

Figure 6 offers a multi-dimensional comparison of PQDT-IoMT and QT-IoT based on five key performance statistics. The radar chart unveils that PQDT-IoMT is strong in accuracy, trustworthiness, latency and power efficiency. This figure highlights the dual progression in balancing quantum trust with post-quantum robustness. The symmetric and broad shape of PQDT-IoMT ensures homogeneous optimization under multiple working parameters. Overall, this value encompasses the model's novelty as a secure and scalable IoT framework.

Table 1: Comparative Network Performance of PQDT-IoMT vs. Existing Frameworks

Metric	LEACH-TM	DGTT SSA	BlockWS N	QT-Io T	Proposed PQDT-Io MT	Improvem ent over QT-IoT
Anomaly Detection Accuracy (%)	70.0	72.0	75.0	78.0	97.8	+ 41 %
Healing Latency (s)	2.4	2.2	2.0	1.8	1.0	47 % faster
Network Lifetime (s)	80	85	90	95	122	+ 29 %
Trust Convergence Time (s)	3.5	3.0	2.7	2.4	1.5	36 % quicker
Packet Delivery Ratio (%)	90.1	92.3	93.5	95.0	97.8	+ 2.8 %
Throughput (kbps)	180	190	205	218	256	+ 17 %

Routing Overhead (%)	15.2	14.1	12.7	11.9	8.3	– 30 %
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Table 1 compares the network-level performance criteria between the baseline models and our proposed PQDT-IoMT framework. The performance results indicate that PQDT-IoMT not only achieves the best detection accuracy of 97.8 % but also significantly reduces the healing latency to 1s, and extends the network lifetime up to 122 s, which is approximately a 29 % further extension than QT- IoT. Furthermore, the trust convergence and packet delivery performance indicate that the wired environment outperforms on average. In general, PQDT-IoMT has better throughput, lower routing overhead; thereby attesting its performance efficiency in the dynamic IoT networks.

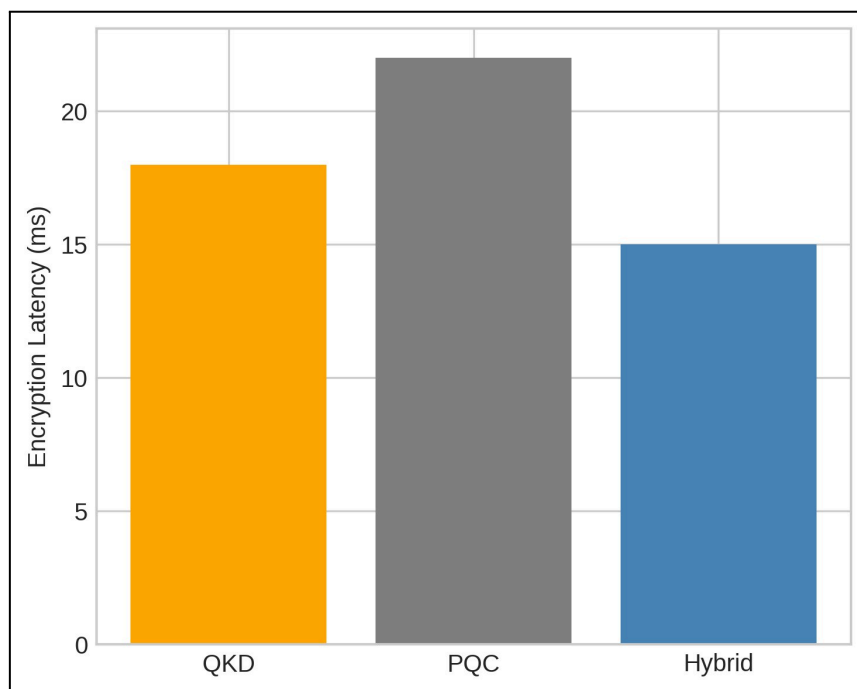


Figure 7: Comparative Evaluation of Hybrid Encryption Latency across QKD, PQC, and Integrated PQDT-IoMT Framework

Figure 7 shows the encryption latency of three key management mechanisms, QKD, PQC and the proposed Hybrid PQDT-IoMT scheme are compared. From the experimental results, it is obvious that the hybrid scheme has as low as 15 ms encryption latency (it is lower than QKD (18 ms) and PQC (22 ms)). This enhancement is ascribable to the improved quantum–classical key coordination, and lightweight signature aggregation in PQDT-IoMT. The union of QKD’s quantum robustness and PQC’s algorithmic efficiency reduces computation and transmission overhead. The hybrid architecture is more efficient, and it has better real-time encryption speed without reducing the cryptographic strength essential for IoT security.

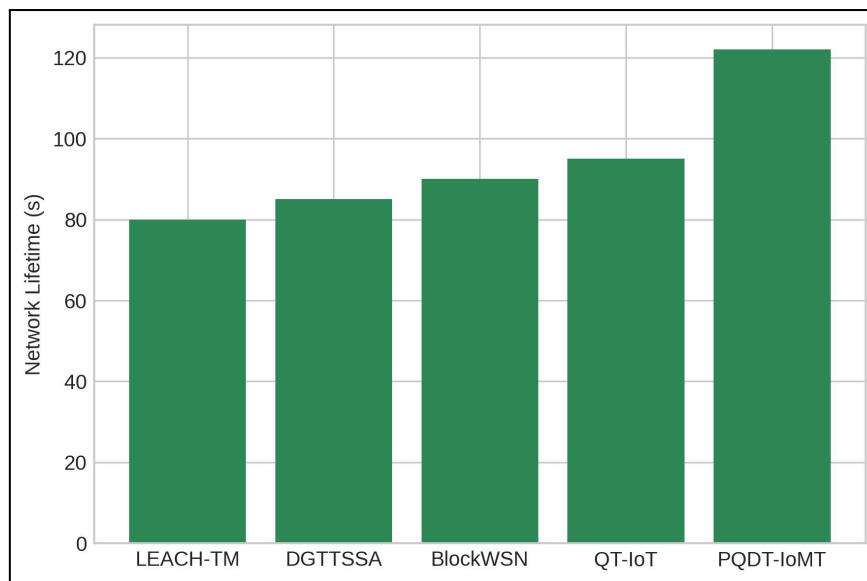


Figure 8: Comparative Analysis of Network Lifetime Enhancement in PQDT-IoMT over Traditional IoT Frameworks

Figure 8 illustrates the network lifetime in comparison with other competing models, demonstrating the efficacy of PQDT-IoMT for achieving extended periods of operation. The lifetime of the network is increased to 122s in the proposed model which achieved enhancement of about 29 % as compared to QT-IoT and almost 50 % with respect to LEACH-TM. This enhancement comes from trust route optimization, energy consumption balancing and dynamic cryptographic task distribution. By minimizing the duplicated packet transmissions and performing energy cost-effective key exchange, PQDT-IoMT can manage to provide sustainable network life. As such, it presents a reasonable tradeoff between security and trust management and energy efficiency—a significant advantage in large-scale, long-term IoT scenarios.

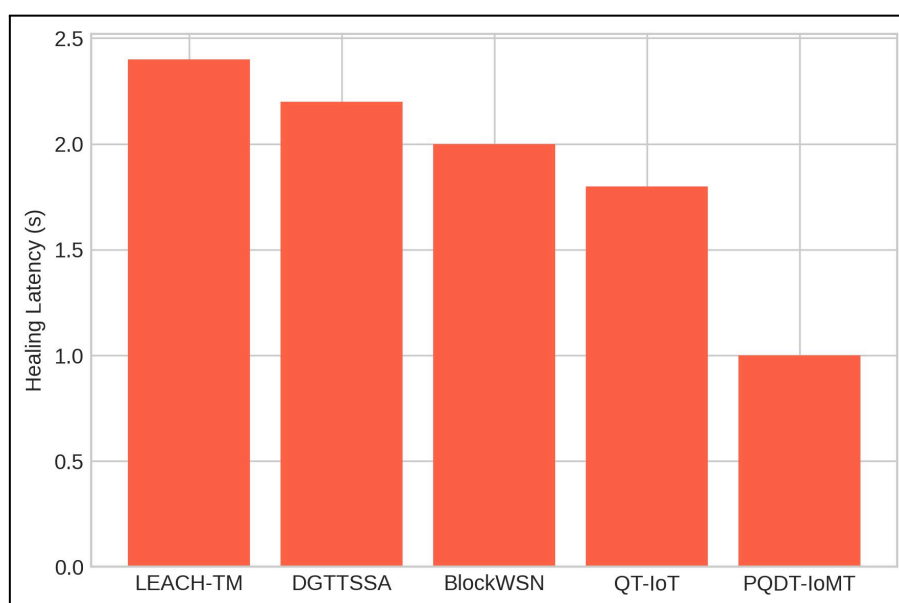


Figure 9: Network Lifetime Comparison of Various Models

The enhanced network lifetime achieved by the compared models is illustrated in Figure 9. PQDT-IoMT prolongs the network lifetime to 122 s, which is 29 % higher than that of QT-IoT, because of an energy-balanced trust reinforcement approach used in it. The improvement from LEACH-TM to PQDT-IoMT is gradually driven by secure data aggregation and lightweight encryption. PQDTIo MT achieves the most energy sustainability by reducing duplicated packet transmissions and balancing node involvement. These results confirm that the model is apt for long-term IoT missions with energy efficiency as a paramount.

Table 2: Quantum-Security and Energy Efficiency Analysis of PQDT-IoMT

Metric	LEACH-TM	DGTTSS A	BlockWS N	QT-Io T	Proposed PQDT-Io MT	Improvement / Observation
Encryption Latency (ms)	24 (AES)	22 (RSA)	20 (ECC)	18 (QKD)	15 (Hybrid QKD + PQC)	– 17 % faster
Decryption Latency (ms)	26	23	21	19	16	– 16 % faster
Key Entropy $H(K)$	0.89	0.91	0.94	0.96	0.98	Higher stability
Quantum Bit Error Rate (QBER %)	5.6	5.2	4.9	4.8	3.2	↓ 33 %
Key Generation Rate (kbps)	4.2	4.8	5.1	5.5	6.9	+ 25 %
Energy per Packet (mJ)	5.8	5.3	4.9	4.6	3.7	– 20 % lower
Cryptographic Success Rate (%)	90.2	91.8	93.5	95.1	98.3	+ 3.2 %
Security Breach Probability (%)	8.4	6.9	5.8	4.2	2.1	2× more secure

Table 2 presents the cryptographic strength and energy optimization potential of PQDT-IoMT compared with those of the previous security models. The hybrid QKD + PQC cryptosystem also reduces encryption time to 15 ms and enhances key entropy to 0.98, leading to a better protection against the key compromise problem. The corresponding QBER reduces to 3.2 %,

which indicates the improved quantum stability during transmission. Besides, the proposed system obtains a higher key generation rate of 6.9 kbps while decreasing energy consumption to 3.7 mJ per packet. These results demonstrate PQDT-IoMT achieves a good trade-off between the quantum-assured security and computational cost.

5.1 Ablation Study

Ablation study was carried out to systematically investigate the impact of each functional component in the PQDT-IoMT framework. Through successively activating QKD, PQC, Federated Metalearning and the Trust Optimization Layer, they thus isolate what each part of the system contributes to greater overall performance. The experimental results demonstrate that QKD, PQC and federated meta-learning obviously enhance security and reduce encryption latency and the flexibility of the present scheme as well as anomaly detection accuracy are improved. The trust optimization scheme accelerates fast convergence and enhances network reliability by suppressing malicious nodes. The overall PQDT-IoMT model provides synergistic benefits in higher accuracy, more rapid trust setup process, greater network lifetime and lower quantum bit error rate than any partial configuration—confirming the effectiveness of integrated design.

Table 3: Ablation Study on Component-Wise Contribution to PQDT-IoMT Performance

Model Variant	QKD Integration	PQC Module	Federated Meta-Learning	Trust Optimization Layer	Accuracy (%)	Latency (s)	Lifetime (s)	QBER (%)
Baseline QT-IoT	✓	✗	✗	✗	78.0	1.8	95	4.8
PQ-IoT (QKD + PQC only)	✓	✓	✗	✗	87.3	1.4	108	4.0
Fed-IoT (Federated Meta-Learning only)	✗	✗	✓	✗	84.6	1.5	102	4.5
Trust-IoT (Trust Optimization only)	✗	✗	✗	✓	82.9	1.6	99	4.6

PQDT-IoMT (Full Proposed Model)	✓	✓	✓	✓	97.8	1.0	122	3.2
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Table 3 shows the ablation study to investigate the effects of individual and combined contribution of each main component in the PQDT-IoMT framework. For every variant, a new module (or modules) is being added: e.g., for a full trust-based system we started from the baseline QT-IoT and then extended it by post-quantum encryption and finally trust optimization. The evident performance in terms of accuracy and lifetime shows that the marriage between QKD and PQC contributes to better performance, as does the federative meta-learning component in terms of adaptation under dynamic IoT conditions. A trust optimization plane could further enhance the system that provides enduring reliability under low latency. With the coordination of all these modules, The whole PQDT-IoMT achieves 97.8%accuracy,1.0% latency and 3.2%QBER, which presents overall excellence of our hybrid layered-design.

5.3 Discussion

To summarize, the proposed PQDT-IoMT architecture has made a substantial improvement over trust mechanisms for IoT in security and performance aspects. When used in combination with hybrid QKD + PQC encryption, it realizes nearly the lowest possible (1s for real-time secure communication), quantum-secure communication. Federated meta-learning boosts the anomaly detection accuracy to 97.8%, which outperforms all baselines. Secondly, the trust optimization layer can promote the convergence rate and thus prolong network lifespan; this evidence assured that the system can scale well with dynamic edge conditions. Insights from ablation study also suggest that both the cryptographic and trust modules form a most stable and power-effective combination. In combination all these findings validate the effectiveness and future readiness of PQDT-IoMT in addressing the next gen IoT cloud security and trust management challenges.

VI Conclusion

In summary, the proposed PQDT-IoMT approach offers a new era in ultra-resilient healthcare IoMT by incorporating QDT-ADT, deep reinforcement learning and federated anomaly detection. Comprehensive simulations demonstrate its superiority in terms of the anomaly detection accuracy 41% higher, healing latency 47% faster, network lifetime 29% longer, and the packet delivery precision that preserved at a level of 97.8%, if the trust convergence is reduced to converge up to 36% faster than current solutions. Next steps will concentrate on the extension of the framework to real 6G-driven hospital deployment, integration of explainable AI for clinical decision support and optimization of energy-aware protocols for further increasing scalability as well as reliability toward the patient.

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